

Second-Hand Book Exchange System :

1. USER Entity

Primary Key: User_ID

Attributes:

- User_ID (PK)
- Name
- Email
- Phone_No
- Address
- Password
- Registration_Date

Relationship:

- A **User** **creates** one or more **Book Listings**. (*1 : M*)
- A **User** **sends** one or more **Exchange Requests**. (*1 : M*)
- A **User** **participates in** one or more **Transactions**. (*1 : M*)

2. BOOK LISTING Entity

Primary Key: Listing_ID

Attributes:

- Listing_ID (PK)
- Listing_Date
- Availability_Status

Relationship:

- Each **Book Listing** is **created by one User**.
- Each **Book Listing** **contains one Book**. (*1 : 1*)
- One **Book Listing** can **receive many Exchange Requests**. (*1 : M*)

3. BOOK Entity

Primary Key: Book_ID

Attributes:

- Book_ID (PK)
- Title
- Author
- ISBN
- Edition
- Condition
- Price
- Description
- Status

Relationship:

- Each **Book** appears in one **Book Listing**. (*1 : 1*)
- Many **Books** belong to one **Category**. (*M : 1*)

4. CATEGORY Entity

Primary Key: Category_ID

Attributes:

- Category_ID (PK)
- Category_Name
- Description

Relationship:

- One **Category** contains many **Books**. (*1 : M*)

5. EXCHANGE REQUEST Entity

Primary Key: Request_ID

Attributes:

- Request_ID (PK)
- Request_Date
- Request_Status

- Message

Relationship:

- One **User** can send many **Exchange Requests**.
- One **Book Listing** can receive many **Exchange Requests**.
- One accepted **Exchange Request** results in one **Transaction**. (1 : 1)

6. TRANSACTION Entity

Primary Key: Transaction_ID

Attributes:

- Transaction_ID (PK)
- Transaction_Date
- Transaction_Type
- Status

Relationship:

- One **Transaction** is created from one **Exchange Request**.
- One **Transaction** has one **Payment**. (1 : 1)
- One **User** can participate in many **Transactions**.

7. PAYMENT Entity

Primary Key: Payment_ID

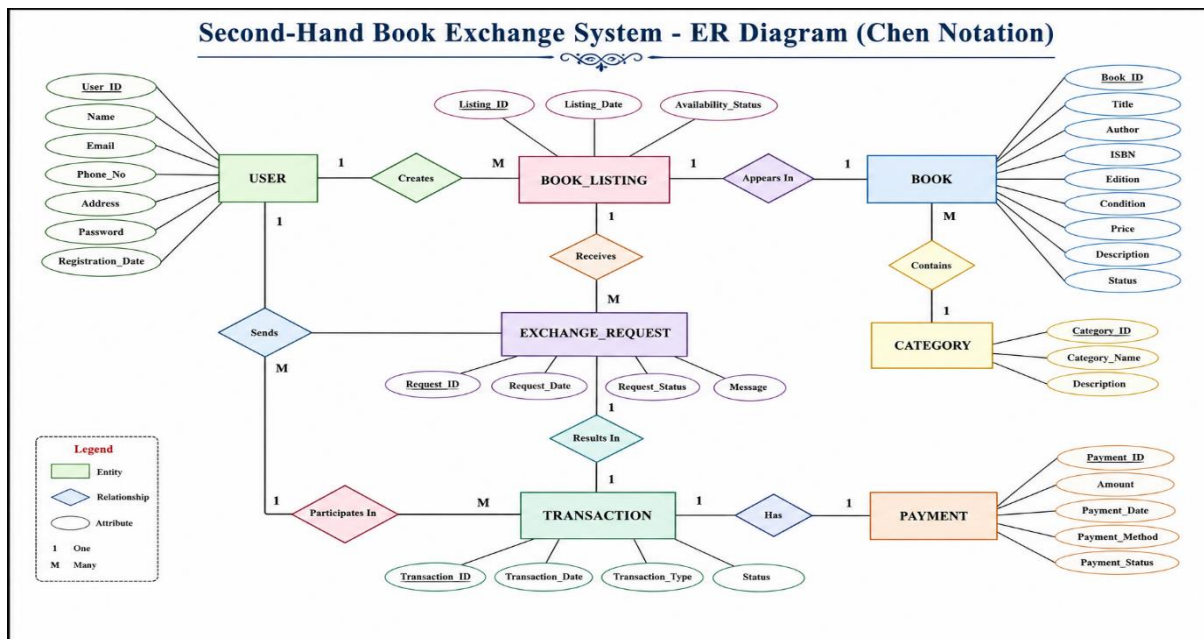
Attributes:

- Payment_ID (PK)
- Amount
- Payment_Date
- Payment_Method
- Payment_Status

Relationship:

- One **Payment** belongs to one **Transaction**. (1 : 1)

ER Relationship model:



Relationship Summary

Relationship	Cardinality
User creates Book Listing	1 : M
Book appears in Book Listing	1 : 1
Category contains Book	1 : M
User sends Exchange Request	1 : M
Book Listing receives Exchange Request	1 : M
Exchange Request results in Transaction	1 : 1
User participates in Transaction	1 : M
Transaction has Payment	1 : 1